Swastik Aditya Ranjan

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LinkedIn | GitHub

Education

National Institute of Technology, Rourkela *August 2023 – Present*
B.Tech in Electrical Engineering | CGPA: 7.74 | Rourkela, Odisha

Odisha Adarsha Vidyalaya, Machhara, Koraput *2020 & 2022*
CBSE Class XII, Science (PCMB) — 91.2% | CBSE Class X — 92.4%

Relevant Coursework: *Control Systems, Advanced Control Systems, Signal Processing, Embedded Systems, Analog Electronics, Power Electronics, Digital Electronics*

Robotics Leadership

Captain & Engineering Lead — Team Tiburon (AUV), NIT Rourkela *Oct 2024 – Present*
EKF/PID/LQR — RP2350 — KiCad — Embedded C — Python — Documentation

- Led design of a BlueROV2 Heavy-class AUV to Finalist standing at SAUVC 2026 (Sanya, China), directing a multidisciplinary team across mechanical, electronics, firmware, and controls.
- Built a 9-state Extended Kalman Filter fusing VN-200 IMU and Teledyne DVL data with pressure-sensor depth for full 6-DOF state estimation.
- Implemented cascaded PID control and derived a weighted pseudoinverse thrust allocator handling reaction-torque and gyroscopic coupling across an 8-thruster configuration.
- Designed the power distribution and debug boards, and documented root-cause resolution of firmware, PID-coupling, and sensor-synchronization failures.

Experience

e-Yantra Summer Internship — Drone R&D *May 2025 – July 2025*
IIT Bombay — Certificate

- Built an indoor autonomous drone using an optical flow sensor and 1D LiDAR for positional stability in GPS-denied environments.
- Integrated and benchmarked ArduPilot, Betaflight, INAV, and PX4 flight control stacks across 150 test flights to determine optimal latency and accuracy.

Projects

Autonomous Drone Navigation *Aug 2024 – Mar 2025*
ROS2 — Path Planning — PID — Python

- Built an indoor autonomous quadcopter using an overhead camera for global positioning and obstacle detection.
- Implemented A* path planning and PID-based waypoint tracking in ROS2.

AXI4 IP Core for GPS–IMU Sensor Fusion *March 2026*
Verilog — Vivado — Zynq-7000 SoC — Kalman Filtering — GitHub

- Engineered a hardware-accelerated GPS–IMU sensor fusion IP core using a Kalman filter with Zero-Velocity Update (ZUPT), interfaced to an ARM Cortex-A9 host via AXI4-Lite and AXI4-Stream.
- Validated filter accuracy and end-to-end data flow through simulation and on-board hardware testing.

STM32H743 Flight Controller Design *Aug 2025 – Nov 2025*
KiCad — 4-Layer PCB — Embedded UAV Hardware

- Designed a 4-layer STM32H7-based flight controller integrating dual IMUs, a BMP581 barometer, an ESP32-S3 wireless module, ideal-diode power management (LTC4413), buck regulation (TPS54531), USB-C, MicroSD, and expansion interfaces.

Technical Skills

Controls & Estimation: Extended Kalman Filter, Cascaded PID, LQR, Thrust Allocation, State Estimation, MATLAB, Simulink

Robotics & Software: ROS2, Gazebo, A* Path Planning, Python, Embedded C, C++

Hardware: PCB Design & Assembly (KiCad, Altium), Microcontrollers (RP2350, ESP32, STM32), FPGA (PYNQ-Z2), Power Electronics

Sensors: IMU (VN-200), DVL (Teledyne), Pressure Transducers, Optical Flow, 1D LiDAR, GPS

Achievements

- **SAUVC 2026 (Singapore AUV Challenge) Finalist**, Sanya, China — among 52 university teams from 16 countries. Certificate
- **Rank 9, NIDAR 2025** — National Innovation Challenge for Drone Application & Research. Certificate
- **e-Yantra 2025 Warehouse Drone Finalist (Rank 5)**, IIT Bombay.
- **Rank 3, Aquatech Innovator Hackathon**, IIT Guwahati — proposed and simulated a soft-robotic bio-inspired AUV for inspection and rescue operations.